

Macroeconomics I — MPE Insper

Mock Final Exam 2 (Simulado 2) — 2026, 3rd term

Instructions. Four blocks of 25 points (100 in total). Duration: 3 hours, closed book, non-programmable calculator allowed. Each block has one objective question (10 points) and one discursive question (15 points). *Multiple choice* (items a, b): mark the single correct option; no penalty for a wrong answer. *True/False* (items c, d): the score of the T/F items is $\max\{0, \text{correct} - \text{wrong}/2\} \times \text{item value}$, i.e. every two wrong items cancel one correct item; blank items do not count as wrong. *Discursive questions*: show every step; box final answers; sign every derivative before interpreting it; label axes and curves in diagrams. Notation: Kurlat (2020) in Blocks I–III, Benigno (2015) in Block IV. All numbers are illustrative.

Block I — Measurement and national accounts

(Lecture 1 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) A soy chain in Mato Grosso, one year, values in R\$ million. A fertiliser producer, which uses no intermediate inputs, sells its whole output, 30, to a soy farm. The farm harvests soy worth 100: it sells 70 to a crushing plant and exports 30 to China. The crusher sells soybean oil worth 150 to Brazilian households. By the production (value-added) approach, the chain's contribution to Brazil's GDP is:

- A. 150
- B. 280
- C. 180
- D. 210
- E. 250

b) (3 points) IBGE data for a year, in R\$ billion: GDP is 10,000; wages earned by Brazilian residents on short assignments abroad, 50; profits of foreign-owned plants in Brazil sent to their parent companies, 300; interest on Brazilian bonds held by non-residents, 150; personal transfers sent home by Brazilians who live permanently abroad, 40. Using Kurlat's definition, $\text{GNI} = \text{GDP} + F^{\text{out}} - F^{\text{in}}$, where F^{out} is factor income earned abroad by residents and F^{in} factor income earned at home by non-residents, Brazil's GNI is:

- A. 9,640
- B. 10,400
- C. 9,750
- D. 9,600
- E. 10,000

c) (2 points, T or F) In a year in which Brazil's gross fixed capital formation equals 17% of GDP, Brazil's stock of fixed capital rises by 17% of GDP.

d) (2 points, T or F) In 2015 Brazil's nominal GDP grew by about 3.8% while real GDP fell by about 3.5%. Hence the GDP deflator rose by less than 3.8% that year.

Question 2. (15 points) Brazil and Portugal: exchange rates, prices and welfare. A think tank compares living standards in Brazil and Portugal with a two-good economy. Per-capita annual quantities and local prices are:

	Traded good T (grains, manufactures)		Non-traded good N (housing, haircuts, services)	
	quantity	price	quantity	price
Brazil	5	10 reais	10	4 reais
Portugal	10	2 euros	15	2 euros

The traded good obeys the law of one price at the market exchange rate. Output per worker is $A_T = 2$, $A_N = 5$ in Brazil and $A_T = A_N = 5$ in Portugal; within each country labour moves freely between sectors and firms are competitive.

a) (5 points) Find the market exchange rate (reais per euro), Brazil's GDP per capita in euros at the market rate and at PPP (Brazil's quantities valued at Portuguese prices), the Brazil/Portugal ratio under each conversion, the PPP exchange rate and Brazil's price level relative to Portugal's.

Solution:

b) (5 points) Which good drives the gap between the two ratios? Derive the relative price of the non-traded good in each country from the firms' and workers' conditions, explain the direction of the bias of the market-rate comparison (Balassa–Samuelson), and explain why “transport and trade costs” is *not* the explanation here.

Solution:

c) (5 points) Welfare beyond GDP. Use Kurlat's consumption-equivalent (Jones–Klenow) measure with log utility ($\sigma = 1$), lognormal consumption within each country and equal leisure (set the leisure term to zero). A person lives in a country with probability $e = \text{LE}/100$ and then enjoys $\bar{u} + \mathbb{E}[\ln c]$. Data: Brazil's mean consumption per capita is 60% of Portugal's (normalise Portugal's to 1); the standard deviation of log consumption is $s_{BR} = 0.9$ and $s_{PT} = 0.5$; life expectancy is 76 and 81 years; $\bar{u} = 5$. Find λ , the factor applied to Portuguese consumption that makes a person behind the veil indifferent between the two countries, decompose $\ln \lambda$ into consumption, inequality and life-expectancy terms, and compare with (a).

Solution:

Block II — Long-run growth and the Solow model

(Lectures 2–3 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) Solow model in *per-worker* terms, no technological progress, $Y = K^{1/3}L^{2/3}$, $s = 0.20$, $\delta = 0.05$. Starting from its steady state, Brazil's population growth falls permanently from 2% to 1% a year (the demographic transition). Which statement about steady-state output per worker y^* , the growth of y during the transition and the growth of aggregate Y in the new steady state is correct?

- A. y^* rises by about 26.0%; $g_y > 0$ only during the transition; g_Y falls from 2% to 1%.
- B. y^* rises by about 8.0%; $g_y > 0$ only during the transition; g_Y stays at 2%.
- C. y^* rises by about 8.0%; g_y is permanently higher than before; g_Y stays at 2%.
- D. y^* rises by about 16.7%; $g_y > 0$ only during the transition; g_Y falls from 2% to 1%.
- E. y^* rises by about 8.0%; $g_y > 0$ only during the transition; g_Y falls from 2% to 1%.

b) (3 points) Same model with $\alpha = 0.35$ and $n + \delta = 0.07$; Brazil saves 17% of GDP. Which statement is correct about the golden-rule saving rate and the effect on steady-state consumption per worker of raising s from 0.17 to 0.25?

- A. $s_{GR} = 0.65$; raising s from 0.17 to 0.25 lowers steady-state consumption per worker.
- B. $s_{GR} = 0.35$; raising s from 0.17 to 0.25 raises steady-state consumption per worker.
- C. $s_{GR} = 0.35$; raising s from 0.17 to 0.25 lowers steady-state consumption per worker.
- D. $s_{GR} = 1.00$; raising s from 0.17 to 0.25 raises steady-state consumption per worker.

c) (2 points, T or F) Of two economies with the same technology, depreciation and population growth, the one with lower capital per worker necessarily grows faster.

d) (2 points, T or F) If a country's national accounts show a capital share of income of one third, the same as in undistorted economies with $\alpha = 1/3$, then its measured TFP reflects only technology.

Question 4. (15 points) “Custo Brasil” in the growth accounts. Output is $Y = AK_P^\alpha L^{1-\alpha}$ with $\alpha = 0.4$, where K_P is productive capital. Regulation, legal uncertainty and crime force every firm to hold θ units of *compliance capital* (duplicate tax-filing systems, armoured vehicles, walls, back-up generators) for each unit of productive capital. Compliance capital produces nothing, but the national accounts record it as capital like any other, so measured capital is $K = K_P + \theta K_P$. Capital is rented at r and labour hired at w in competitive markets. Today $\theta = 0.6$.

a) (5 points) Write aggregate output as a function of A, K, L, θ . What share of measured capital is unproductive? How much output is lost relative to an economy with $\theta = 0$ and the same A, K, L ? State the base of your percentage.

Solution:

b) (5 points) Write the representative firm's problem and first-order conditions and derive the

measured factor shares rK/Y and wL/Y . A development accountant has data on Y , K , L and factor payments only. What does she conclude about α and about TFP? Who bears the cost of θ ?

Solution:

c) (5 points) A reform (tax simplification, digital filing, lower crime) cuts θ from 0.6 to 0.28 gradually over the decade 2027–2036, while true A stays constant; K and L grow at whatever rates they grow. Write the growth-accounting decomposition, compute the average annual Solow residual, say what it would show if θ instead *rose* from 0.28 to 0.6, and explain why this is consistent with Kurlat's reading of the residual.

Solution:

Block III — Microfoundations: consumption, labour and general equilibrium

(Lectures 4–6 • 25 points)

Question 5. (10 points) Objective items.

a) (3 points) A two-period household (Kurlat ch. 6) has $u = \ln c_1 + \beta \ln c_2$ with $\beta = 1$, faces $r = 0$, can borrow and lend freely and earns $y_1 = y_2 = 10,000$ reais. Consider three separate scenarios, each against this baseline: (i) a one-off bonus of 1,000 today (an extraordinary FGTS withdrawal); (ii) a credible announcement today of a raise of 1,000 paid in period 2 only; (iii) a permanent raise of 1,000 in both periods. The changes in c_1 in (i), (ii), (iii) are:

- A. 1,000; 0; 1,000
- B. 500; 0; 1,000
- C. 500; 500; 1,000
- D. 500; 500; 500

E. 1,000; 500; 2,000

b) (3 points) The government must raise a given revenue from households that save. It can tax interest income or levy a lump-sum tax in period 2 that raises the same revenue. With CRRA utility, which statement is correct?

- A. Only the tax on interest changes c_2/c_1 , through the net return in the Euler equation; at equal revenue the lump sum gives higher utility.
- B. Both taxes lower c_2/c_1 by the same amount, because they raise the same revenue in present value and so lower wealth equally.
- C. Only the lump-sum tax changes c_2/c_1 , because it lowers wealth; the tax on interest is neutral once its revenue is fixed.
- D. Only the tax on interest changes c_2/c_1 , through the net return in the Euler equation; at equal revenue both give the same utility.

c) (2 points, T or F) A household with $u = \ln c_1 + \beta \ln c_2$ earns $y_1 > 0$ in period 1 and nothing in period 2 (no taxes), so it is a net saver. When r rises it raises c_1 , because for a saver the income effect of a higher return dominates the substitution effect.

d) (2 points, T or F) With $u = \ln c - \chi h^{1+\eta}/(1+\eta)$ and no non-labour income, hours do not respond to a permanent change in the wage (the Marshallian elasticity is zero). Therefore a temporary, anticipated wage rise within a life of given wealth leaves hours unchanged too: the Frisch elasticity is also zero.

Question 6. (15 points) An agribusiness productivity boom in general equilibrium. A one-period economy has a fixed capital stock $\bar{K}$. The representative household owns $\bar{K}$ and the firm, has one unit of time and utility $u = \frac{c^{1-\sigma}}{1-\sigma} + \psi \ln(1-L)$ with $\psi = 2/3$ (σ is the curvature of utility; $1/\sigma$ the elasticity of substitution). The competitive firm produces $Y = A\bar{K}^\alpha L^{1-\alpha}$ with $\alpha = 1/3$. At the baseline $A\bar{K}^\alpha = 2^{2/3}$ and $\sigma = 2$. New seed varieties raise A in agriculture, which we treat as a rise in aggregate A .

a) (5 points) Write the household's and the firm's problems, the first-order conditions and the market-clearing conditions. Show that profits are zero. Reduce the equilibrium to one equation in L , verify that $L = 1/2$ at the baseline, and compute c , w and $r^K \bar{K}$.

Solution:

b) (5 points) Derive $d \ln L / d \ln A$ in general and sign it by σ . Evaluate it at the baseline and give the first-order effects of a 10% rise in A on L , w , r^K and c . Explain in words that match the sign.

Solution:

c) (5 points) Now set $\sigma = 1$. The government taxes labour income at $\tau = 0.25$. Compute equilibrium hours without the tax and in two cases: (i) the revenue is rebated lump sum; (ii) the revenue is thrown away (spent on something with no value to the household). Explain why hours fall in (i) even with log utility, why they fall less in (ii), and what happens in (ii) if $\alpha \rightarrow 0$.

Solution:

Block IV — Money, inflation and the New-Keynesian AS–AD model *(Lectures 7–9)*

• 25 points)

Question 7. (10 points) Objective items.

a) (3 points) A flexible-price economy (Kurlat ch. 11) has real money demand $M/P = Y \ell(i)$, $\ell' < 0$, with Y and r fixed by the real side. Which statement is correct?

- A. A one-off 10% rise in M raises P by 10% and leaves Y , r and M/P unchanged; a permanent rise in money growth also leaves M/P unchanged, as money is superneutral.
- B. A one-off 10% rise in M raises P by 10% and leaves Y , r and M/P unchanged; a permanent rise in money growth raises i and lowers M/P .
- C. A one-off 10% rise in M lowers i through a liquidity effect and raises Y for good; a permanent rise in money growth raises i and lowers M/P .
- D. A one-off 10% rise in M raises P by 10% and leaves Y , r and M/P unchanged; under a Selic target the Copom fixes both i and M independently.

b) (3 points) Real money demand is $M/P = 0.10 Y e^{-4\pi}$ (Cagan form, π in decimal per year), Y is constant and in steady state π equals money growth. The inflation rate that maximises seigniorage, and seigniorage at that rate, are:

- A. $\pi = 12.5\%$; seigniorage 0.76% of GDP
- B. $\pi = 25\%$; seigniorage 2.50% of GDP
- C. $\pi = 50\%$; seigniorage 0.68% of GDP
- D. $\pi = 25\%$; seigniorage 0.92% of GDP

c) (2 points, T or F) In Benigno's AS–AD model, a cut in the Selic moves the economy along the AS curve.

d) (2 points, T or F) In Benigno's model, natural output y_n differs from efficient output y_e because a fraction of prices is sticky.

Question 8. (15 points) A mark-up shock in fuel distribution. A merger wave concentrates fuel distribution and raises firms' desired mark-up by 4.4% for one period only. Use Benigno's model

in log deviations from an efficient steady state, with no public spending ($g = 0$), no tax changes, $\bar{p} = p^e = 0$ and $\bar{y}_n = 0$:

$$\text{AS: } p - p^e = \kappa(y - y_n), \quad \kappa = \frac{(1 - \alpha)(\sigma^{-1} + \eta)}{\alpha}; \quad \text{AD: } y = \bar{y}_n - \sigma[i - (\bar{p} - p) - \rho].$$

Calibration: $\alpha = 0.66$ (share of sticky prices), $\sigma = 0.5$, $\eta = 0.2$, $\theta = 8$, $\rho = 5\%$. Before the shock $i = \rho$, $y = y_n = y_e = 0$ and $p = p^e$. At first the Copom keeps the Selic at 5%.

a) (5 points) Which curve shifts and which does not, and why? Find y and p , and the signs of $y - y_n$ and $y - y_e$. Draw the AS–AD diagram and the time path of y and p over the short run (period 1) and the long run (period 2).

Solution:

b) (5 points) With $u(C) = \frac{C^{1-1/\sigma}}{1-1/\sigma}$, $v(L) = \frac{L^{1+\eta}}{1+\eta}$, $Y = AL$ and $C = Y$, the market's labour-supply condition is $\text{MRS} = A/(1 + \mu)$ and the planner's is $\text{MRS} = A$. Derive both, log-linearise, and show $y_n - y_e = -\mu/(\sigma^{-1} + \eta)$. Explain why sticky prices play no role, and evaluate the gap for this shock.

Solution:

c) (5 points) The Copom minimises $\mathcal{L} = \phi_y(y - y_e)^2 + \phi_p(p - p^e)^2$ with Benigno's welfare weights $\phi_y = \frac{1}{2}$, $\phi_p = \theta/(2\kappa)$, subject to AS. Derive the targeting rule by substitution, the optimal y and p and the split of the shock, and the Selic that implements it. Explain why AS and not AD is the constraint, and what the money market does. Is the optimal response a rise or a cut relative to doing nothing, and on what does that sign depend?

Solution:

End of exam.